INCH-POUND
MIL-M-14/2
AMENDMENT 2
19 May 1993
SUPERSEDING
AMENDMENT 1
7 November 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, PHENOLIC, THERMOSETTING, CONTAINING MINERAL/GLASS FILLERS

This amendment forms a part of MIL-M-14/2, dated 2 April 1990, and is approved for use by all Department and Agencies of the Department of the Defense.

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* Table I: Delete and substitute: (page 2 attached)

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* Table I: Delete and substitute: (page 3 attached)

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Table II: Delete and substitute: (page 4 attached)

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - AR
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Preparing activity:
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Navy - AS, OS
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User activities:
Navy - YD, EC

AMSC N/A

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TABLE I. Qualification test requirements for phenolic resin molding compounds, mineral/glass filled.

Requirement	Type MFE	Type MFH	Type GPG	Type GPI-5	Type GPI-10	Type GPI-20	Type GPI-30	Type GPI-50	Type GPI-100
Mechanical/physical									
Compressive strength, endwise	15,000	15,000	23,000	25,000	25,000	23,000	20,000	20,000	20,000
Flexural strength	8,000	7,000	9,000	12,000	12,000	12,000	12,000	14,000	15,000
Heat deflection temperature	115	130	170	175	175	175	175	175	200
Heat resistance	175	200	175	200	175	175	175	175	175
Impact strength, side[1]	-	0.25	0.3	0.5	1.0	2.0	3.0	5.0	10.0
Tensile strength	4,200	4,200	4,500	7,000	6,500	6,500	6,500	6,000	4,500
Water absorption	0.10	0.35	0.30	0.35	0.35	0.40	0.50	1.0	1.5
Electrical									
Dielectric breakdown:									
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)									
Step-by-step test	45	35	35	35	35	35	35	35	40
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)									
Step-by-step test	40	10	15	15	15	15	15	15	15
Dielectric constant:									
at 1 kilohertz	6.0	-	7.0	7.0	7.0	7.0	7.0	7.0	7.0
	6.0	-	8.0	8.0	8.0	8.0	8.0	8.0	8.0
at 1 megahertz	6.0	-	6.0	6.0	6.0	6.0	6.0	6.0	6.0
	6.0	-	6.3	6.3	6.3	6.3	6.3	6.3	6.3

See footnote at end of table.

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TABLE I. Qualification test requirements for phenolic resin molding compounds, mineral/glass filled - Continued.

Requirement	Type MFE	Type MFH	Type GPG	Type GPI-5	Type GPI-10	Type GPI-20	Type GPI-30	Type GPI-50	Type GPI-100
Electrical (Continued)									
Dielectric strength:									
Short-time test	325	215	275	250	250	250	250	250	300
Step-by-step test	275	150	225	200	200	200	200	200	200
Short-time test	325	125	225	175	175	175	175	175	75
Step-by-step test	275	80	200	125	125	125	125	125	50
Dissipation factor									
at 1 kilohertz	0.03	--	0.08	0.08	0.08	0.08	0.08	0.08	0.08
	0.03	--	0.09	0.09	0.09	0.09	0.09	0.09	0.09
at 1 megahertz	0.15	--	0.05	0.05	0.05	0.05	0.05	0.05	0.05
	0.02	--	0.06	0.06	0.06	0.06	0.06	0.06	0.06
Surface resistance	5.0	--	--	--	--	--	--	--	--
Volume resistance	2.0	--	--	--	--	--	--	--	--
Combustion									
Flame resistance: Units - Seconds (min. avg.) Test method - D 229 See 4.5.2.9									
Ignition time	60	100	100	100	100	100	100	100	100
Burning time	210	200	200	200	200	200	200	200	200
Flammability/Thickness: Inch: Test method UL 94 See 4.5.2.10									
	V-1/1/16	V-0/1/16	V-1/1/16	V-0/1/16	V-1/1/16	V-1/1/16	V-1/1/16	V-1/1/16	V-1/1/16

1/ The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. Batch acceptance test requirements for phenolic resin molding compounds, mineral/glass filled.

Property to be tested	Type MFE	Type MFH	Type GPG	Type GPI-5	Type GPI-10	Type GPI-20	Type GPI-30	Type GPI-50	Type GPI-100
Dielectric constant at 1 megahertz	6.0	--	6.3	6.3	6.3	6.3	6.3	6.3	6.3
Dielectric strength, step-by-step	275	80	200	125	125	125	125	125	50
Dissipation factor at 1 megahertz	0.02	--	0.06	0.06	0.06	0.06	0.06	0.06	0.06
Flexural strength	8,000	7,000	9,000	12,000	12,000	12,000	12,000	14,000	15,000
Impact strength, side[1]	--	0.25	0.3	0.5	1.0	2.0	3.0	5.0	10.0
Water absorption	0.10	0.35	0.30	0.35	0.35	0.40	0.50	1.0	1.5

1/ The side of the test specimen is that area formed by the chase of the mold.

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